# Using LLM Digital Twin Simulation for Synchronous Research: With an Application to the Structured Group Design*

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## Abstract

We use LLM digital twin simulation to model human discursive group interaction under alternative synchronous research design choices, at scale and across repeated trials – revealing the sampling distribution of the qualitative results under each design. These methods generalize to any synchronous discussion design, and enable us to state the assumptions required for the simulation to add to the accumulation of knowledge as well as validation methods that can help weaken these assumptions. The software is open source, and easy to use and modify. We use these simulation methods to show that in repeated trials, the traditional, small-$N$ synchronous design either neglects minority viewpoints, or it generates text data that is uncorrelated across trials, indicating a "garden of forking paths" discussion outcome. In contrast, we evaluate a novel large-$N$ *structured group design* under which text data remains stationary across repeated trials and inferentially captures diverse perspectives and reasoning within populations.



Keywords: Digital Twins Simulation, Large Language Models, Algorithmic Curation, Synchronous Design, Deliberation, Focus Groups

Word Count: 9,363

# 1 Introduction

Human opinions are formed through social interactions (Chambers, 2018; Habermas, 1984). Learning how public opinion emerges from social interactions requires synchronous research designs that collect closed- and open-ended data while participants actively engage each other (Esterling et al., 2021). Engaging in synchronous group-based interaction can lead participants to access deeper cognitive processes (Esterling et al., 2011; Tetlock, 1983) rather than just offering top of the head responses like in a standard survey (Zaller and Feldman, 1992).[1] The design or context for those interactions, however, heavily conditions the nature and dynamics of discussion (Cyr, 2019; Dryzek et al., 2019; Fishkin, 2009; Niemeyer et al., 2024), and so there is a need to understand the *properties* of alternative discussion designs for evaluating emergent group interaction.

An ideal method for evaluating the properties of discussion designs would require hosting a discussion and collecting data using a specific experimental design many times over, across different designs, each time with human participants. This procedure would reveal the sampling distribution of results generated under each design. Using human participants in synchronous designs, however, is extremely expensive and time consuming – particularly when the design scales to require hundreds or thousands of participants that must meet at the same place and time.[2] Here we develop low-cost methods for digital twins LLM simulation to explore the emergent properties of group discussion under different discursive contexts and designs, and at arbitrary scale (Wang et al., 2025; Zhang et al., 2024). We specify validation tests and state the assumptions required for digital twins simulation methods to contribute to the accumulation of knowledge for discursive design.

[1]Asynchronous platforms cannot replicate the dynamics of synchronous human discursive interaction as they tend to reduce spontaneity, shorten answers, and create uneven discussion flow and unfocused exchanges compared to synchronous platforms (Zhang et al., 2024).

[2]To give a sense of the costs, the "Connecting to Congress" Deliberative Town Hall experiments (Neblo et al., 2018) used probability sampling to recruit participants for online synchronous meetings, and each session of 20-30 participants cost about $10K to host. The AmericaSpeaks "Our Budget, Our Economy" synchronous in-person deliberation event (Esterling et al., 2021) recruited over 2,000 participants, and the project required a year of planning and a budget of over $2M.

The digital twins methods we introduce are general to any discursive setting. Here we introduce and evaluate the *structured group design*, which is a novel design for human synchronous discussion groups that can scale to any size group. The structured group design enables synchronous human participant input at scale through a three-stage discussion format centered on a substantive topic, motivated by a closed-ended survey question, relying on an LLM pipeline to capture diverse perspectives submitted simultaneously by all participants in real time during a synchronous event (similar to Lu et al., 2024). Our project contributes to an emerging literature on using LLMs to develop deliberative systems and platforms (Estornell and Liu, 2024; Friess and Eilders, 2015; Karacapilidis et al., 2024; Ma et al., 2024; Rountree et al., 2026; Tessler et al., 2024; Zhang et al., 2024). The software to implement the structured group design – with either human or LLM-simulated participants – is open source, easy to use, and adaptable to any research topic or question.

The structured group design contrasts with traditional synchronous designs where only one participant can speak at a time, such as deliberation experiments (Esterling et al., 2021; Neblo et al., 2018) or focus groups (Cyr, 2019). In the traditional setting, the discussion group must remain small so that each participants has a chance to speak. The small group design, however, is vulnerable to undesirable group dynamics, including *unrepresentativeness*, *path dependence* based on the specific individuals present and the speaking order, and *group think* where the conversation is dominated by a given perspective (Neblo et al., 2019). These vulnerabilities limit the ability of synchronous group results to generalize inferentially to a larger population and to produce meaningful or valid discursive data. These vulnerabilities are remedied in the structured group design.[3]

We use digital twins LLM simulation methods (Rountree et al., 2026) to explore the properties and behaviors of the structured group under alternative design choices. The simulation shows, among other things, that the small-$N$ design is vulnerable to at least two pathologies. First, the small-$N$ design often neglects disfavored or unpopular view-

[3]Of course, the structured group design is not appropriate if the opinion-formation hypothesis involves face-to-face interaction, such as the study of conflict aversion or emotions in discourse.

points, and second, among the viewpoints discussed, the qualitative text data collected is poorly correlated across replications, indicating that conversations in a small group follow an idiosyncratic "garden of forking paths" logic where the synchronous group conversation does not systematically capture opinions within a population. In contrast, the large-$N$ design robustly surfaces viewpoints supported by even small minorities of participants, and explores the full conversation space that exists in a larger population. We identify the assumptions required in order for the digital twins simulation to contribute inferentially toward the accumulation of knowledge regarding discursive designs, and we show validation methods that can help to weaken those assumptions.

## 2 Synchronous Design and Inferential Research

The synchronous group design enables the researcher to collect more nuanced data on considered opinions than can be captured in a standard survey. The group-based discussion replicates the kinds of everyday social interactions that generate opinions and can reveal meanings and understandings from the participant's own (emic) perspective (Cyr, 2019; Habermas, 1984), revealing not just what opinions and beliefs individuals hold but why they hold them and how they are formed (Chambers, 2018; Fishkin, 2018).

We posit that discovering the systematic discussion on a topic or issue within a population, or that would occur within a population, is a core goal of synchronous group research. As Fishkin (2018) notes, the synchronous group design should create a data generating process where the discussion data that results from the group matches or reflects the discussion that would occur within a larger population, had the larger population engaged in a similar discursive effort. Unfortunately, the traditional design for the synchronous group is unlikely to yield data that recovers this population-level perspective. In the traditional design, the number of participants must remain small – generally between six and eight participants per session – so that all participants have an opportunity to speak

(Cyr, 2019). The small-$N$ design creates vulnerabilities, however, that make the results likely not generalizable or representative of an emic, interpersonal understanding among members of a population for the issue under discussion (see Neblo et al., 2019).

Specifically, small group dynamics make the discussion within the small group heavily dependent on the specific individuals present (Parker and Tritter, 2006). If the individuals who happen to be present are not sufficiently diverse, it is possible that some perspectives in the population are never represented or expressed in the discussion – and a small-$N$ inherently cannot encompass the perspectives of diverse populations (Boas, 2024; Conover and Searing, 2005; Leung and Savithiri, 2009). In addition, discussion within a small group where each person speaks in turn is likely to create a path dependence based on who is talking and the order in which they speak (Cyr, 2019). The conversation will navigate idiosyncratically down one path within a garden of forking paths and not engage in a full exploration of the conversation space (Neblo et al., 2019). Finally, small groups are vulnerable to "group think" where the conversation is dominated by one specific perspective (Parker and Tritter, 2006; Sunstein, 2002), although this concern can be mitigated with effective discussion moderation (Esterling, 2018).

The unrepresentativeness of participants coupled with path dependence and group think limit exploration of the full discussion space in a target population.[4] A preferred design will ensure diverse viewpoints are expressed, and ensure that discussion is not path dependent and will not be vulnerable to group think.

# 3 The Structured Group Design

We propose an LLM-assisted research design for a synchronous meeting that can scale to any $N$, which we call the *structured group* design. The user interface design and algorithmic curation pipeline to implement the structured group design for human participants

[4]In much of the focus group methods literature, authors assert that synchronous research is only useful for exploratory research. Our methods are able to extend synchronous methods to inferential research.

has been implemented in the collabinar platform called *Prytaneum*,[5] a platform that is open source, web-based and free to use.[6] Scaling the synchronous group to a large $N$ enables inferential research under relatively weak assumptions that we state below, and so addresses many of the limitations with the traditional synchronous group design. These assumptions can be further weakened with validation methods. In addition, the structured design, by design, reduces the risk of group think within a discussion, even in the case when the group remains small.

The structured group design involves three stages, for each question of interest. In the first stage, the facilitator first provides background information and framing, the ground rules and motivation, and then presents a question to the audience with a set of closed-ended response options paired with a text box requiring participants to explain the reasoning why they chose their preferred option. For example, in our primary simulation below, we use the question, "What is your preferred approach to developing nuclear power?" With response options 1: "Implement rapid expansion and investment of nuclear power;" 2: "Maintain the current nuclear energy operations without change;" 3: "Phase out existing nuclear plants and halt new construction;" 4: "Prioritize researching improved implementations of nuclear power." Of course, the question and standpoint options can be anything the researcher chooses. Once the facilitator closes the poll, the collabinar algorithm generates a summary of the participants' open-ended reasoning to support each of the four options. If none of the respondents chooses a given option there will be nothing to summarize and the summary for that option will be empty.

In the second stage, the facilitator shares the summarized reasonings with the structured group participants, and then asks the participants to read a reason supporting one of the standpoint options they oppose (among the options that have a non-empty sum-

[5]https://prytaneum.io/

[6]The structured group design can be implemented as an asynchronous survey using three waves. However, the asynchronous survey design would lack a facilitator and discussion lead, and would likely experience significant attrition across the three waves, and suffer from the limitations of asynchronous platforms such as reduced spontaneity, shorter answers, and uneven discussion flow and unfocused exchanges compared to synchronous platforms (Zhang et al., 2024).

mary), and then to write a counterargument to that reasoning.[7] Once the second stage ends the algorithm summarizes the counterarguments submitted for each of the non-empty options. Then in the third stage, the facilitator asks participants to read the summarized counterarguments to the preferred option they selected in stage 1 and then asks them to write a response to the counterargument. The third stage responses are then summarized by the algorithm. Each of the summaries, along with all of the individual responses and all measured covariates, are stored in a database for post-session analysis.

The structured group design scales to any number of participants. Irrespective of the number of other participants, each participant has a robust opportunity to express their views regarding the question under discussion. Assuming a large number of participants, and provided the options are well-constructed, there should be sufficient reasoning to support each of the options to reflect the full distribution of perspectives in the underlying population of interest from which the participants are drawn, analogous to the concept of inclusive discursive representation in deliberation theory (Dryzek, 1990). Importantly, since each participant is asked to consider and respond to alternative perspectives, the design reduces the risk of group think (Sunstein, 2000). Further, with a large $N$, the discussion summaries should inferentially capture systematic features of the underlying discussion, and hence should be stable over repeated experiments. And finally, the full diversity and nuances of individual perspectives are captured in the responses across three stages of discussion that are stored in the database and can be mined for further insights.

In contrast, small-$N$ groups under the structured design can yield final summaries that are inconsistent across experiments or exclude reasoning to support the options favored by unpopular or underrepresented perspectives. And if the small-$N$ design shows these limitations under the structured group design, there is no reason to believe that switching to an unstructured traditional synchronous group design would remedy those limitations, and indeed would simply re-expose the discussion to the risk of group think.

[7]Alternatively, the researcher can randomly assign an opposing viewpoint in the second stage. A complete randomization ensures that each non-empty option will receive counterarguments.

## 4 Digital Twin Simulation

We propose a general method to simulate the results of synchronous human interaction in order to understand the properties of alternative research designs. Specifically, we use LLM prompting to simulate the discursive data generation process using alternative design choices for the structured group design. These simulations allow us to explore the properties of different sample sizes, under differing sampling methods and response options, in repeated trials, at considerably lower cost and more efficiently than if we recruited humans to generate the data.[8] To improve the simulation methods, we make use of a digital twins approach. Digital twins technology uses data from the physical world in order to anchor the simulation to the physical process being modeled, provided the simulation undergoes sufficient validation (National Academy of Sciences, 2024).

There is considerable work on digital twin silicon sampling for survey research (Arora et al., 2025; Brand et al., 2023; Chuang et al., 2024; Dillion et al., 2023; Hämäläinen et al., 2023; Korst et al., 2025; Sarstedt et al., 2024), along with several projects that curate databases to power a digital twins pipeline (Hu et al., 2025; Park et al., 2024; Toubia et al., 2025). To create personas, we pipe digital twin covariate data into the LLM prompt, creating instructions for the persona the LLM is to adopt, as well as background information and the previous stages of the discussion as context which improves consistency of LLM performance (Zhang et al., 2024).

As with all LLM-based technologies, digital twin methods must be carefully validated before they can be relied on to add to the accumulation of knowledge (Bail, 2024; Bisbee et al., 2024; Estornell and Liu, 2024; Messeri and Crockett, 2024; National Academy of Sciences, 2024). In addition to hallucinations, LLMs are known to exhibit systemic biases inherited from the training data (Blodgett et al., 2020; Borkan et al., 2019; Caliskan et al., 2017; Feng et al., 2023; Hofmann et al., 2024; Hu et al., 2024; Motoki et al., 2024; Sorensen et al., 2024; Taubenfeld et al., 2024; Tjuatja et al., 2024) or from alignment processes

[8] The simulations in this paper cost about $20 total and require about two days to run.

(Santurkar et al., 2023). Digital twins and silicon sampling methods can introduce other distortions. For example, when prompted to respond from the perspective of a specific persona, LLMs can flatten or misrepresent the perspectives of identity groups (Sarstedt et al., 2024; Wang et al., 2025). LLMs also perform less well in representing non-English speaking populations (Qu and Wang, 2024).[9]

To validate the simulations for our proposed structured group design, we evaluate the output of the ensemble of LLM personas to ensure that the LLMs are correctly responding to the prompt instructions – that the LLM is adopting the instructed persona and responding appropriately to the initial survey question and to the other LLM inputs. We also evaluate the summaries generated at the conclusion of each stage to ensure that they correctly capture the diverse reasoning in a systematic and unbiased way.

To evaluate the LLM persona output, we make use of three tests, relying on the framework for "algorithmic fidelity" proposed by Argyle and her collaborators (2023). The first test is *backward continuity*, which requires that the LLM responses are consistent with the digital twin persona. To test backward continuity, we mask the digital twin persona instructions that were included in the prompt and ask a research assistant to infer the covariate values from the LLM generated text. The second test is *frontward continuity*, which tests whether the LLM output is a sensible response to the prompt. To test for frontward continuity, we ask a research assistant to read the prompt and assess whether the generated text is a natural or sensible response to the prompt.[10] The third test is *coverage*, which is to compare the generated responses among the LLMs to a known list of arguments that have been generated by humans on the corresponding topic, and evaluate whether the LLMs express the full diversity of human viewpoints on the topic.

[9]In principle, LLM output can be debiased using regularization, fine-tuning, or alignment processes such as reinforcement learning from human feedback (Agiza et al., 2024; Bang et al., 2021; Bolukbasi et al., 2016; Dixon et al., 2018; Jin et al., 2021; Lin et al., 2024; Liu et al., 2020, 2021). Debiasing would be difficult or impossible to implement with human participants.

[10]Frontward continuity stands as an informal Turing test (Aher et al., 2022), which requires that LLMs generate text indistinguishable from human generated text. A formal Turing test would randomize between LLM and human responses and ask the research assistant to guess which response is human.

We also assess the extent of bias in the abstractive summarization methods (Altemeyer et al., 2025; Mittelstadt et al., 2023; Zhang et al., 2020). As we describe below, we use cosine similarity to measure the closeness of each persona's response to the corresponding summary and evaluate whether distances from the summary are random or if they are systematically related to the demographic covariate values (Wang et al., 2025).

# 5 Simulation for the Structured Group Design

Here we make use of the digital twins survey database provided by Toubia et al. (2025). In the simulations, we sample from the 2,058 Toubia survey respondents and use the covariate information to create a unique digital persona via prompt engineering, requesting the LLM to adopt the persona reflected in the digital twin covariates in order to provide open- and closed-ended survey responses. Since the respondents in the Toubia digital twin database are representative of the U.S population, random samples from the database provide covariate information that reflects the joint distribution in the U.S. population.

We created simulated structured group data by using a digital twins approach to design an ensemble of LLM personas – that is, a set of LLMs prompted to have diverse views driven by covariates that reflect the U.S. population. Using the structured (multi-round) group protocol described above, we have the LLM personas engage repeatedly in trials under alternative design options, with different topics and survey prompts. To assess the consistency and stability of responses across trials, we calculated pairwise cosine similarity comparisons for the resulting text generated within each of the response summaries, for each standpoint option, across repeated simulated structured group trials under each design. To calculate the cosine similarities, we first convert the text output to embedding vectors which represent the semantic meaning of the relevant comparison text.[11]

[11] Cosine similarity is a measure of the semantic similarity between two strings of text that is widely used in natural language processing. To calculate cosine similarity, we use the Hugging Face sentence transformer `all-MiniLM-L6-v2` to map each text response into a 384 dimension vector, and then apply the cosine rule. Cosine similarity scores range from zero (orthogonal text) to one (identical text).

The cosine similarity of the summaries between two trials evaluates the stability of the summarized reasoning for each option expressed across repeated trials. If a synchronous design inferentially explores the full conversation space for a topic, then the summaries across trials will be semantically similar resulting in high cosine similarity scores on average with low variation across trials. If the design results in idiosyncratic discussions then the summaries across trials will be dissimilar, resulting in low cosine similarity scores across trials and high variance. However, stability across trials by itself does not demonstrate an inference to a population-level discussion. This inference requires stability to be combined with two assumptions, coverage and unbiasedness, that we discuss below.

The question and standpoint option choices for the structured design discussion are selected by the researcher. For example, below we focus on the topics of nuclear power, gun control and increasing the size of the U.S. House of Representatives. For each prompt we specify parameters to determine the "persona" of the LLM participant using digital twin covariates that explain variation in attitudes toward the topic, along with an additional parameter that decides which LLM to use for that persona.[12] For example, in our simulation on nuclear power, the covariates are *education-level* (less than college degree, college or more), *gender* (male, female), *political ideology* (liberal, conservative), and *environmentalism* (pro-environmental movement, not pro-environmental movement). The corresponding prompt for nuclear power for stage 1 is:

> You will adopt the personality of a [education-level]-educated, [gender], [ideology]-leaning participant of a focus group discussing the adoption of nuclear power. You have [level of affinity] for environmentalism. You must select from one of the four possible standpoints on nuclear power and briefly provide your reasoning for doing so in 1-2 sentences.[13] The possible standpoints are:

[12]We diversify the LLMs so the results are not confined to a specific LLM. We used 4 LLMs: Gemini 3.0 flash preview, Qwen3 32B, Llama 3.3 70B versatile, and OpenAI GPT OSS 120B, with the latter three being open weight LLMs. Each persona used the same LLM model and covariate values across the three rounds of the focus group.

[13]The prompt also provides background information as context to help the LLM model personalities, The background we use is taken from this PEW report: https://www.pewresearch.org/short-reads/2025/10/16/support-for-expanding-nuclear-power-is-up-in-both-parties-since-2020/. "To help you choose an option, please know that according to PEW Research Center, about half of US residents favor expanding nuclear power, while the other half prefer to maintain operations without change or to phase

- 1: Implement rapid expansion and investment of nuclear power.
- 2: Maintain the current nuclear energy operations without change.
- 3: Phase out existing nuclear plants and halt new construction.
- 4: Prioritize researching improved implementations of nuclear power.

After the personas respond, we request Google Gemini to summarize the reasons offered for each standpoint option (among non-empty options) using the following prompt:

> You are a helpful assistant tasked with summarizing argumentative statements on the topic of nuclear power. I have gathered a list of statements that participants in an online focus group platform have provided as their reasoning to [standpoint option text]. Summarize the following statements as no more than five concise summaries. Your summaries should be worded as passive tone viewpoints, so that they capture as many viewpoints presented in the given statements as possible.

The LLM's responses in stage 1 create three inputs to the second stage: *stage1-standpoint* (the participant's chosen standpoint from stage 1), *stage1-reasoning* (the participant's reasoning for choosing stage1-standpoint), and *opposing-viewpoint* (a randomly selected summarized reason to support a standpoint that the persona did not choose in stage 1). The prompt for stage 2 is:

> You will adopt the personality of a [education-level]-educated, [gender], [ideology]-leaning participant of a focus group discussing the adoption of nuclear power. You have [level of affinity] for environmentalism. You have previously stated that your standpoint is to [stage1-standpoint], and your reasoning for this was: '[stage1-reasoning]'. As part of the focus group discussion, a different viewpoint has been brought up: '[opposing-viewpoint]'. You must now provide a counterargument for this viewpoint on nuclear power in 1-2 sentences.

The responses from stage 2 are summarized in a manner similar to the summarizations in stage 1, which creates one more input to the third stage, *counterargument* (a summarized response from stage 2 that argues against this participant's original standpoint). The prompt for stage 3 is:

it out. Most men favor expanding it while most women prefer phasing it out or keeping it the same. Republicans are more likely to favor expanding compared Democrats. Environmentalists have mixed opinions, where some favor expanding it because nuclear has a low-carbon footprint, while others favor contracting it because they are concerned about the risks of nuclear waste disposal."

> You will adopt the personality of a [education-level]-educated, [gender], [ideology]-leaning participant of a focus group discussing the adoption of nuclear power. You have [level of affinity] for environmentalism. You have previously stated that your standpoint is to [stage1-standpoint], and your reasoning for this was: '[stage1-reasoning]'. As part of the focus group discussion, a counter-argument to your standpoint has been brought up by a participant who said that: '[counterargument]'. You must now respond to this counterargument in 1-2 sentences.

The responses in stage 3 are summarized again within each (non-empty) standpoint.

The simulation then saves the persona-level responses for each stage along with all inputs such as the covariates defining the persona, the opposing standpoint used in stage 2, and the counterargument used in stage 3. It also saves the summaries from each round. For illustration, the appendix provides the summarized responses for the three stages for a single $N$=60 group on nuclear power, showing the arguments raised in the structured group. The project repository houses all of the simulated output across all trials we report. As separate repository distributes the Python scripts needed to run the structured group simulator; one can use the scripts to replicate the findings of this paper, or to run simulations on any topic and question of interest.

# 6 Results under Random Sampling

We conducted sets of digital twins simulations on the topics of nuclear power, gun control, and increasing the size of the U.S. House of Representatives. The nuclear power simulations reported in this section use simple random sampling of personas to establish the random baseline sampling distribution for the structured group design under alternative design choices. Researchers who use the traditional, small-$N$ synchronous design often use purposeful or stratified sampling in order to ensure perspective diversity within discussion groups. To explore the consequences of stratified sampling, in the next section we use the highly polarized topic of gun control and stratify groups by liberal-conservative ideology. Finally, we use the simulated nuclear power data and human data on expanding the House of Representatives in order to conduct validation tests and replications.

We start with the nuclear power issue and use simulation to discover the sampling distribution of the structured group design under random sampling. Making use of the structured group design, we ran 50 trials of the small-$N$ version with six participants each; 30 trials of the medium-$N$ version with 60 participants each; and 20 trials of the large-$N$ version with 600 participants each. Each "participant" was one of the randomly selected digital twin LLM personas.

Then for each standpoint, we create an $N \times N$ matrix of pairwise cosine similarities, measuring the similarity of the summary of stage 3 responses for each trial with the summary of stage 3 responses for every other trial within the standpoint and within that design. We retain only the lower triangle of this matrix, excluding the diagonal. If a summary is empty for a given standpoint on a given trial, the cosine similarity with every other trial is zero by definition.[14] For the $N = 6$ participant version trials we have 1,125 pairwise comparisons; for the $N = 60$ trials we have 435 comparisons; for the $N = 600$ trials we have 190 comparisons.[15]

Comparing these similarity scores across our nuclear power trials shows the stability of responses within a given structured group design under random sampling, within each standpoint option. Figure 1 compares the densities of the across-trial cosine similarity scores for designs with six, 60 and 600 personas, conditional on the response option. The within-response distributions of the cosine similarity scores show that the small-$N$ version with size six has a mode at zero for each of the standpoints, indicating that in many trials, no persona advocated these standpoint options (the summary was empty) with some frequency.[16] This mode at zero is small, however, for the fourth standpoint

[14] If the summaries are empty for two trials, we set the cosine similarity to zero.

[15] Each 600 participant simulation takes about an hour to run. Fortunately, since the results are nearly identical across trials, we do not need to run many trials to recover the large-$N$ sampling distribution and 190 trials is a sufficient number.

[16] In our simulations, we can analytically recover the rate that a standpoint is excluded. For example, the option to phase out nuclear power was only selected by 7.7 percent of personas in stage 1 in our simulation, and hence we can expect that 68.1 percent of groups with $N = 6$ will exclude this standpoint. If one were to make the (very strong) distribution match assumption that we discuss below, then one

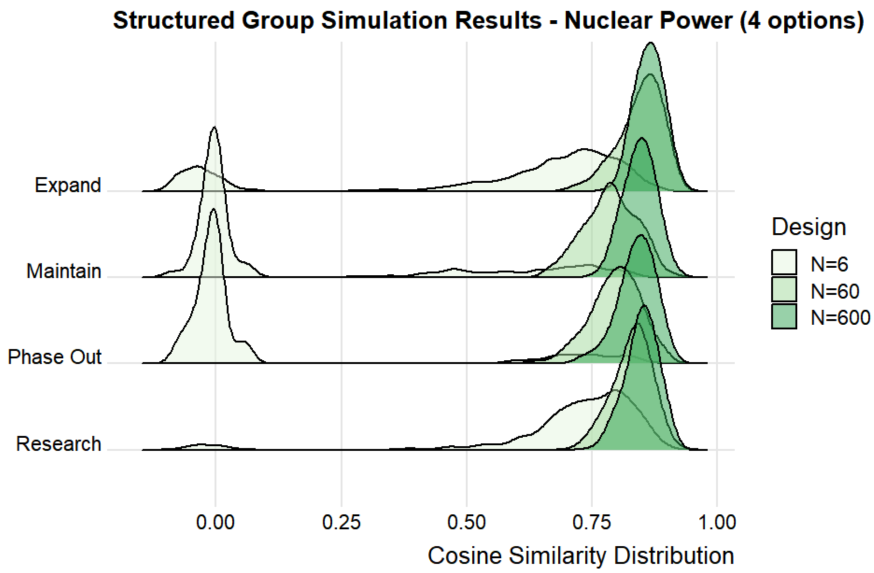


Figure 1: Distribution of pairwise cosine similarity scores for stage 3 summaries across trials, comparing between the small-$N$ group design and the medium- and large-$N$ design under random sampling. Simulated structured group results on the topic of expanding nuclear power. Response 1: Implement rapid expansion and investment of nuclear power. Response 2: Maintain the current nuclear energy operations without change. Response 3: Phase out existing nuclear plants and halt new construction. Response 4: Prioritize researching improved implementations of nuclear power.

option which was to prefer researching improved implementations of nuclear power.

This fourth option was the most popular among all LLM personas, but as a result, for groups with only six participants, that meant that for many trials there were no participants that favored one or more of the other options, and hence these other options were never even discussed in the structured group. In addition, when an option is discussed in a small-$N$ group design, the cosine similarities indicate wide instability in the response across trials with non-zero modes centered at about 0.7 and wide dispersion – showing an interaction under the small-$N$ design between the (in)frequency that an option is discussed and the idiosyncrasy of the discussion.

could use the simulation to estimate the rate by which each standpoint is excluded from the discussion. To warrant the distribution match assumption, one would need to include the optimal covariate set and background information in the persona prompt in order to match the distribution of choices among the standpoints to match the distribution in a target population. Optimizing to warrant the distribution match assumption is an ongoing area of research and is outside of the scope of this paper.

By comparison, the 600 participant focus groups have a unimodal distribution with relatively low variance across trials centered at about 0.9. Conceptually, one can think of the sequence of experiments in the small-$N$ design is one that results in a multimodal distribution of divergent results while the sequence of experiments in the large-$N$ design remains a stable, unimodal stationary distribution. And importantly, the degree of instability of the small-$N$ design itself varies across the response options, while the large-$N$ design returns results that are always stable and of similar magnitude across the response options, irrespective of the relative popularity of each option. The performance of the trials with 60 participants is between these two extremes.

This simulation shows limitations in the traditional small-$N$ group design, and the merits of the large-$N$ design. Conceptually, just as with quantitative research, we see a strong instability of results across trials for the small-$N$ design. This means that the results for the small-$N$ design are not inferentially reliable representations of the underlying discussion that is often the object of discovery in synchronous group research. Instead, small-$N$ group dynamics create qualitative data that either reflects idiosyncratic discussion, or excludes unpopular viewpoints in the population. In contrast, the large-$N$ design yields a draw from a stationary distribution that reflects a systematic summary of the discussion that does or could occur in the population. We can take these summaries from the large-$N$ design as inferential to a population-level discourse if this stability is combined with two assumptions, coverage and unbiasedness, which we discuss below.

Notice that the "popular" standpoint option to research the implementation of nuclear power is unlike the other three options. That is, the other three options are to expand, to maintain the same level, or restrict nuclear power, which are regulatory and are mutually exclusive and exhaustive. A vast majority of the LLM personas were attracted to the non-regulatory research option for whatever reason and hence the distributions of responses were skewed in our trials. We repeated the same sets of trials of six, 60 and 600 participants, but this time excluding the option to research nuclear power from the

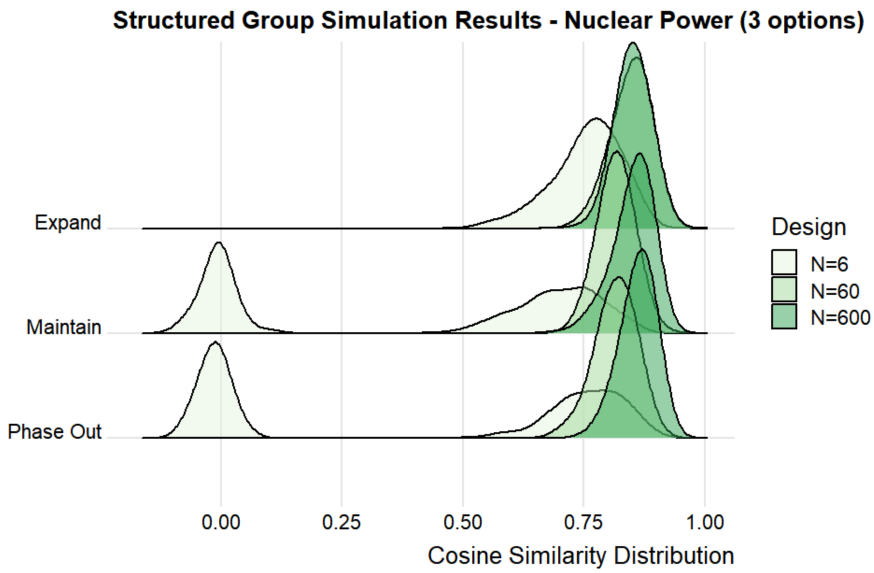


Figure 2: Distribution of pairwise cosine similarity scores, comparing between the small-$N$ group design and the large-$N$ design – three response options. Simulated focus group results on the topic of expanding nuclear power. Response 1: Implement rapid expansion and investment of nuclear power. Response 2: Maintain the current nuclear energy operations without change. Response 3: Phase out existing nuclear plants and halt new construction.

choice set which yields response distributions that are less skewed.

The results of these trials are shown in figure 2. Note that when the response distribution is relatively unskewed, the performance of the 60 participant design nearly matches that of the 600 person design. This indicates that a larger sample size is required in the presence of viewpoints supported by small minorities of participants, and improved instrument design can reduce the $N$ needed for reliable results. This simulation approach to understanding the sampling distribution of the data is analogous to recent proposals to use simulation for power analysis under complex research designs (Blair et al., 2019). These results also show a possible tradeoff in discursive research, where placing constraints on the discussion can increase the stability of the results.

# 7 Stratified Sampling Illustration

Given that a small sample of six or eight participants will have a considerable amount of sampling variability, synchronous group researchers understand that stratified or even purposeful sampling can improve on simple random sampling to ensure diversity in a given group (Cyr, 2019). For example, researchers interested in methods of ideological depolarization will compose small groups that balance liberals and conservatives (Esterling et al., 2021). We are able to use our digital twin simulation methods to explore benefits to small group research that can occur under such stratified sampling.

Of course, stratification does not help when covariates only weakly explain positions on topics such as nuclear power, but it might help improve the balanced representation of perspectives for topics where the stratifying covariates predict responses. Here we explore the consequences of stratification on self-reported ideology on the issue of gun control.[17] We build the LLM personas using the same gender, education, ideology and model covariates we describe above, and run trials for $N = 6$ where we stratify sampling to ensure that half of the participants are liberal and half are conservative, $N = 6$ with simple random sampling, and $N = 60$ with simple random sampling. This setup allows us to compare the stratified selection design to the two random baselines.

We set three standpoint options for the LLM personas to choose from: "Gun control laws should be more strict," "Gun control laws are about right," and "Gun control laws should be less strict." For context, we include the background information text: "To help you choose an option, please know that according to the PEW Research Center, 82 percent of U.S. conservatives believe that gun control laws should be the same or less strict while 92 percent of U.S. liberals favor making gun control laws more strict. That is, liberals favor making gun control laws more strict and conservatives to keep the laws the same or make them less strict."

[17] For background, see the PEW report https://www.pewresearch.org/short-reads/2024/07/24/key-facts-about-americans-and-guns/

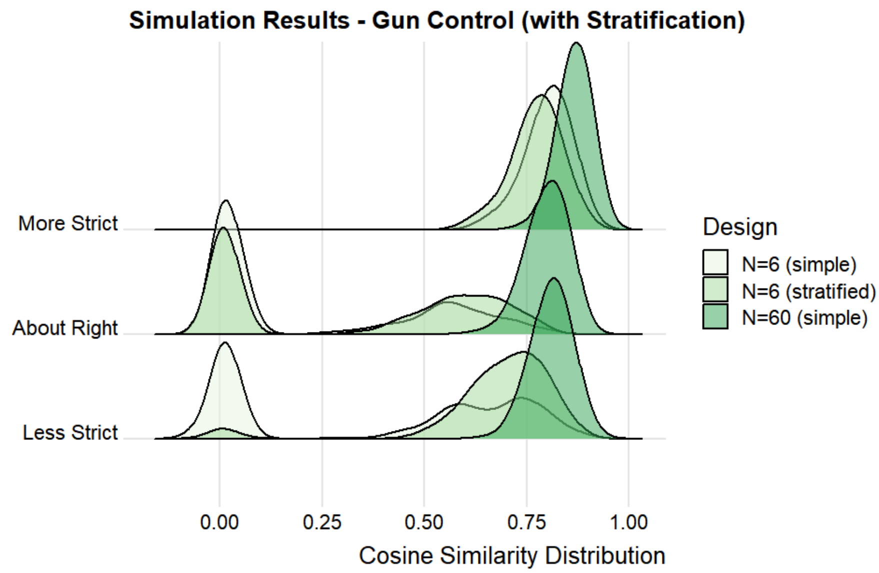


Figure 3: Simulated focus group results on the topic of gun control. Distribution of pairwise cosine similarity scores, comparing between the small-$N$ group design and the large-$N$ design – three response options with stratified sampling comparison to two random baselines. Response 1: Gun control laws should be more strict. Response 2: Gun control laws are about right. Response 3: Gun control laws should be less strict.

Figure 3 shows the results. Compared to the small-$N$ simple random sample design, stratification by ideology does help to ensure balance across the two polarized response options, to either be more restrictive or less restrictive, but it has no effect on the more neutral option to advocate that gun control laws are currently the right amount of regulation. The results show that this stratification design is effective if the purpose of the discussion is to ensure participants are speaking across differences, but less effective if the purpose of the discussion is to inferentially discover the full underlying discussion space (again, under the coverage and unbiasedness assumptions we discuss next).

# 8 Assumptions and Validation

The extent to which digital twin simulation can be relied on to add to the accumulation of knowledge depends on a set of application-specific assumptions (Bail, 2024; Bisbee et al., 2024; Estornell and Liu, 2024; Messeri and Crockett, 2024; National Academy of Sciences, 2024). While it is common for researchers to articulate assumptions for validity in statistical applications (Esterling et al., 2025), to date the LLM literature has not developed norms to state assumptions required for simulation research.

Our application of digital twins simulation to evaluate synchronous group design relies on two core assumptions: 1) *Coverage:* with enough participants, the LLM-based discussion recovers the full range of human-made arguments within a given population on the topic of discussion across the three stages, or that could be made, and 2) *Unbiasedness:* that the discussion summaries do not systematically exclude the perspectives of identity groups. Given these assumptions, we can draw inferences regarding the discussion space in the underlying population. These are relatively weak assumptions that can be further weakened by our validation tests.

Importantly, note that there are two further, and relatively strong, assumptions that would be required in the ordinary case of survey research but that we do not need. These are 3) *Faithfulness:* that the digital twin personalities produce choices, reasoning or content that reflects the (intersectional) demographics of the digital twin covariates. It is well known that the documents in the LLM training text data are not associated with demographic covariates, and hence LLMs are not well-designed to have the LLM output match the voice of identity groups defined by covariates (Wang et al., 2025). In our application, we use digital twins prompting to ensure that the arguments that LLMs voice in the group discussion ecologically capture the diversity of human argumentation, but our application does not require the argument content at the persona level to correspond to the demographic identity group's voice reflected in the covariates.

We also do not require the assumption 4) *Distribution match:* that the digital twin personalities select the standpoint options in a way that matches the marginal distribution of those preferences in the target human population (Bisbee et al., 2024), either conditioned or unconditioned on covariates. Our structured group design is motivated by the notion of discursive representation (Dryzek, 1990), and hence we only need to have a viewpoint expressed by at least one LLM persona for that perspective to be included in the corresponding summary. The marginal distribution of preferences among the participants is not strongly relevant to discursive representation.[18]

We rely on validation tests to weaken our two required assumptions, coverage and unbiasedness. We use our backward continuity, frontward continuity and coverage validation tests to evaluate the coverage of human-made arguments in the discussion, and we rely on a bias audit to evaluate whether our summaries capture systematic perspectives.

To assess the LLM coverage of human arguments, we first consider the backward continuity test, where we masked the covariate information and asked a research assistant to infer the covariate values based on the generated text alone for 60 observations on the topic of nuclear power. This test checks if the LLM is following the prompt instructions to create a given persona. We rely on $\chi^2$ tests from confusion matrices. We find that the research assistant is 26 percent more likely to correctly infer female rather than male ($p = 0.063$), 31 percent more likely to correctly infer college educated rather than not college educated ($p = 0.022$) and 25 percent more likely to correctly infer liberal rather than conservative ($p = 0.036$). In addition, the true environmentalism score is about a standard deviation higher when the research assistant inferred the persona was positive toward environmentalism ($p < 0.001$). Finally, we also masked which standpoint option the LLM chose, and the research assistant was able to infer the standpoint chosen from the text of the reasoning offered and made the correct inference for 58 out of the 60 responses. Note that backward continuity weakens both the coverage and the faithfulness assumptions, although in our application we only need it to weaken the former.

Second, we consider the frontward continuity test, where we asked the research assistant to read the prompt created for each persona and to evaluate if the LLM output was

[18] For example, if 100 LLM personas offer an identical perspective, and only one LLM persona expressed a different perspective, then the summary would include both perspectives. The marginal distribution of participants holding a given perspective is not strongly relevant.

responsive to the text in the prompt. This is similar to a Turing test. Here the research assistant was asked to read all three rounds for a 60 participant trial. Out of the 180 opportunities for the LLM to be non-responsive, the research assistant found only five cases where the LLM did not respond to the prompt. However, when the lab team conducted a review of those five responses, we found the output to be responsive in each case. By design, our pipeline does not depend on the memory of the LLM across the three rounds – instead, we pipe in the text from each previous round into the prompt. This method avoids the limitations of LLMs' diminished performance in multi-turn conversations, and as a result the LLMs are responsive on each API call. One concern might be that LLMs are even more responsive to the prompt text than humans would be, creating a hyper responsiveness that would not be present in human-generated text.[19]

Third, we implement validation tests for issue coverage. Here we make use of an existing reference listing of pro and con arguments regarding nuclear power from the *Encyclopedia Britannica*,[20] which provides three pro arguments that nuclear power: 1) has zero emissions and can reliably and inexpensively support an electricity grid; 2) is a perfect complement to weather-dependent renewable energies; and 3) protects American global national security interests; along with three con arguments that nuclear power: 4) is dangerous, from the risk of meltdowns to the problem of nuclear waste; 5) is expensive to initiate and delays implementation of truly sustainable and renewable energy sources; and 6) is inextricably linked to lethal nuclear weapons.

We provided both a human coder and an LLM-based coder a candidate list of 60 persona-generated reasons for and against nuclear power and asked each coder to decide if each candidate reason matches with an argument on the Britannica reference list, and if so, which one. The two coders disagreed on 22 out of the 60 classifications. The lab team reviewed each instance of disagreement and found that seven disagreements were cases where the candidate reason expressed both reference arguments (so both coders correct),

[19]This is equivalent to filtering human response data to include only on-topic, substantive responses.
[20]https://www.britannica.com/procon/nuclear-power-debate

the human coder made 11 errors and the LLM coder made four errors.

Using the LLM-coded classifications, in this session of 60 participants, the first four of the *Britannica* arguments were expressed in the first round of discussion but the last two con arguments, arguments 5 and 6, were not. However, argument 5, that investing in nuclear power diverts funding from renewable sources, was raised repeatedly in the counterarguments and responses to counterarguments in stages two and three of the discussion. And, it is reasonable to argue that the final con arguments, drawing the analogy to nuclear weapons, is a strange and not especially strong argument – so its absence from the discussion is not surprising in retrospect. In addition, the LLM personas expressed eight additional arguments that are not found on this reference list. One can see all of the summarized arguments across the three stages for this trial in the appendix.

Finally, we develop a bias audit method to detect bias in the summarizations. Note that the stable distribution that results from the large-$N$ design is both normatively and inferentially defensible only if the qualitative summaries of the responses are an unbiased reflection of the individual contributions made during the meeting. If the AI-based summary tool is biased, then the summaries that result would be objectionable in that they would systematically exclude certain perspectives. It is well-known that the underlying LLMs that drive AI applications can have biases inherited from their training data or from alignment interventions (Borkan et al., 2019; Caliskan et al., 2017; Kambhatla et al., 2022; Liu et al., 2020; Santurkar et al., 2023). Thus it is important to conduct an audit of the LLM output to verify the results are unbiased.

We conduct this audit in two ways, first using the simulated data from our experiments which we report in the appendix, and then using a human-generated dataset which we report in the next section.

# 9 Bias Audit with Human Data

To evaluate bias in the LLM summaries, we make use of a human dataset that we were provided that links open-ended response text to demographic covariates. The data come from an online Deliberative Town Hall (Neblo et al., 2018) held by the Institute for Democratic Engagement and Accountability (IDEA) at the Ohio State University.[21] The town hall discussion centered on topics related to the modernization of Congress, and the participants were a U.S. nationally-representative sample. After the town hall, the participants were asked to respond to a survey that collected demographic information and then they were asked to indicate their support for or opposition to a proposal to expand the size of the U.S. House of Representatives, and they also were asked to provide their reasoning for their position as an open-ended response. This data is very similar to the kind of data that would be collected in a structured group design in that the responses with the reasoning that participants submitted followed from a synchronous town hall discussion where they were exposed to both pros and cons.

There were 586 responses to this question.[22] We use our LLM summary tool to create summaries of the favor and oppose responses, then we calculate the cosine similarity of each (human) individual's reasoning with the summary of the position they chose. We ask the summary tool to include five different perspectives within each summary. We then calculate the cosine similarity between the individual's response and each of the five perspectives for their standpoint, and then retain the maximum value. That is, if individual $i$ was in favor of expanding the size of the House, we calculated the cosine similarity between their response and the five in-favor summarized perspectives, and then saved the maximum of these five values.

Figure 4 shows the distribution of these individual-level cosine similarity scores, conditional on demographic covariates and on support or opposition position. Note that

[21]The data was collected under the Ohio State University IRB protocol 2019B0075.

[22]We dropped responses with fewer than 10 characters.

the cosine similarity is higher for the support than the opposition position, which simply indicates that there are more diverse ways to express opposition than support. But conditional on position, the distributions do not show strong substantive, systemic disparities across the demographic categories. The view of college-educated individuals who are in favor of the reform are slightly over-represented, which is not a surprise given that LLMs tend to be biased in favor of well educated responses (Santurkar et al., 2023).

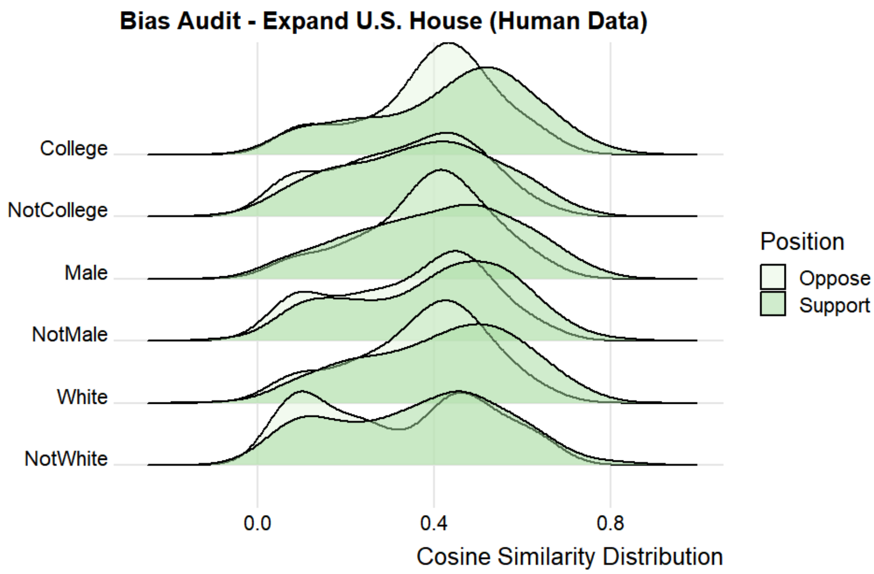


Figure 4: Distribution of cosine similarity scores, conditional on demographics and position. Here, cosine similarity is between 1) an individual's open-ended response and 2) the summary corresponding to the individual's chosen standpoint. Higher cosine similarity indicates the summary captures the individual's perspective, while lower cosine similarity indicates the summary excludes the individual's perspective. Topic: Expansion of the U.S. House of Representatives

Importantly, certain views of non-white respondents who are opposed to the reform seem to be slightly under-represented, given the mildly bimodal conditional distribution, which indicates that some of the perspectives in this group are not well-captured in the summaries. However, the bias audit helps to automate the detection of excluded perspectives since it is straightforward to surface the perspectives of those in the mode. We manually inspected the responses that composed this mode of the conditional distribution and found in each case, the semantic perspective was captured in the summary, but not the surface-level grammar of the expressed response such as in the example, "Too many cooks spoil the soup." And of course, also in contrast to the traditional synchronous group design, our method allows for this kind of bias audit to detect systemic disparities, and the AI-based summarizer can be debiased so that the summaries are orthogonal to demographic characteristics in the event that they do fail an audit (Bolukbasi et al., 2016; Dixon et al., 2018; Liu et al., 2020). In a traditional group design, where individuals speak in turn, there is no method to detect or to correct these kinds of systemic biases that occur in real time during a free-flowing discussion.

# 10 Inference to Population-Level Discourse

The high variance of cosine similarity scores for the summaries across trials for the small-$N$ design indicates that each discussion was in some sense idiosyncratic, and hence shows that the results can not be taken as generalizable inferences to a population-level discussion. In contrast, the stability of the cosine similarity scores for the large-$N$ design suggests that the summaries were capturing perspectives on the topic systematically. However, this stability by itself does not justify an inference to discourse in a larger population. To make this inference deductively requires the coverage assumption to ensure that all perspectives are included, and the unbiasedness assumption to ensure that the summary does not exclude perspectives. All scientific deductions rely on assumptions (Esterling et al., 2025), irrespective of the epistemology of the method. We propose coverage and unbiaseness as minimum assumptions necessary for the simulation to contribute to the accumulation of knowledge in synchronous design research.

# 11 Demonstrating the Generality of the Methods

The project GitHub repository (URL omitted for review) contains all of the open source Python scripts needed to run the digital twin simulator for the structured group design, along with detailed instructions. To run the simulations, one simply needs to download the scripts, modify the topic, standpoint options and background information to a topic and question of interest, and run the script locally. One can leave the scripts at their default values to replicate the results we report.

The methods we describe in this paper can apply to any synchronous research design. However, to assess a new design, a new Python script for the simulator would need to be written. The author team collaborated to write the existing Python scripts by hand for each of the simulations and each of the validation tests we have described so far. However, to show the ease by which these methods can be applied to new designs, we used Claude Code to create a simulator for what we call the *unstructured group design*. We requested Claude to build the unstructured group design script by first examining the existing structured design simulator script, and then modifying the structured design script by removing the closed-ended response options, and instead only have the personas brainstorm ideas on the topic through a multi-round discussion. The goal of this experiment was to show generality of our simulation methods by having Claude modify the script to implement a different synchronous design given all of the dependencies in code.

The new script for the unstructured design simulator only required two requests from Claude, an initial request and then a simple correction, and it took only minutes to complete. The unstructured version worked first time off the shelf.[23]

The unstructured design simulator is available at our project GitHub repository. More generally, this synchronous digital twin simulation project is fully open source and the code is accessible at our repository. We encourage anyone who is interested to join our

[23] The possibilities for alternative designs are endless. For example, an extension for the unstructured design could be to use a stance extraction tool to assign opposite-stance reasons across the rounds to create discussion across differences for the unstructured group design.

open source project as a contributor to help build a library of alternative designs.

# 12 Discussion and Conclusion

We propose an LLM-based digital twins pipeline that can be used to evaluate the properties of different synchronous designs. Digital twins simulation enables the researcher to explore the properties of alternative discussion designs computationally at low cost and effort by simulating the sampling distribution of qualitative data under the different designs. These methods enable the researcher to computationally understand the complex, emergent nature of human discussion under a given design, similar to how researchers use computational methods to understand the power of complex statistical models when an analytical solution is unavailable (Blair et al., 2019). These methods can be used to pilot alternative designs prior to investing in human trials. Finally, we state assumptions required for these simulations to add to the deductive accumulation of knowledge for discussion designs, as well as validation tests that can weaken these assumptions.

As a proof of concept, we apply our simulation pipeline to evaluate alternative designs for synchronous group research. The application of our methods using the structured group design helps to show the properties of designs for human interaction. All of these applications use digital twins simulation to demonstrate the clear superiority of our proposed large-$N$ structured group design to that of the traditional small-$N$ design. In the traditional synchronous group design, a small and unrepresentative number of participants speak sequentially, a data generating process that creates path dependence and group think, and excludes unpopular, minority views. This results in qualitative data and inferences that lack generalizability. In contrast, the large-$N$ group design enables a large number of participants to, in effect, speak simultaneously and to understand all of the diverse perspectives among participants, eliminating the problems of path dependence and bias in the data generating process. We also show that stratification can help for cases where responses are well-predicted by covariates.

Further, we propose the coverage and unbiasedness assumptions needed to understand if the simulation can advance the accumulation of knowledge on research design, and validation tests designed to weaken those assumptions. In particular, we show that the LLM summary tool at the heart of our design is inclusive of diverse perspectives and does not seem to be vulnerable to systemic biases that can plague LLM-based tools. The stability of the summarized responses across trials, combined with the coverage and unbiasedness assumptions, allow us to make inferences to population-level discourse and the formation of public opinion.

Of course, the structured group design is not useful or appropriate if the focus of one's research is an investigation of deeply interpersonal human interactions, such as on the role of conflict aversion or emotions in small group settings. Further, the small-$N$ design remains useful as a tool for exploratory qualitative research (Cyr, 2019). But there is no reason to limit synchronous group methodology to exploration. The large-$N$ structured group design is not vulnerable to the problems of unrepresentativeness, path dependence or group think that plagues the traditional small-$N$ group design. Our proposed simulation methods show how synchronous group designs can extend to inferential qualitative research, and can help researchers uncover a nuanced, interpersonal understanding of the merits of alternatives regarding a decision, in a way that meets the normative standards of deliberative theory (Dryzek et al., 2019; Niemeyer et al., 2024).

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# Using LLM Digital Twin Simulation for Synchronous Research: With an Application to the Structured Group Design

## Supplemental Information Appendix

Table of Contents

# A Validation Data

To conduct the validation of the main simulation, we make use of a single trial for a focus group with $N = 60$. In this trial, the LLM personas debated the question of regulating nuclear power that omitted the non-regulatory option to continue researching nuclear power. Since each trial is a random draw from the LLM, it does not matter which trial we investigate – so we simply used the first trial as our data source for the validation. To find the validation data, navigate to the reproduction GitHub repository (URL omitted for review) and you will find the data in the folder with trial id 4260266019. Table A1 provides the text summaries for the three stages of the LLM persona discussion for this trial, where the first stage summarizes the initial reasons given, the second stage summarizes counterarguments to those reasons, and the third stage summarizes the reactions to the counterarguments.

# B Additional Simulation Results

**Simulated Replication of the Ohio State Deliberative Town Hall.** For comparison to the human data results we report in the text, we replicate the Ohio State Deliberative Town Hall that focused on the topic of the expansion of the U.S. House of Representatives as a digital twins simulation using the structured group design. We implement the simulation with two response options: 1) increase the size of the U.S. House of Representatives by adding more seats, and 2) keep the size of the U.S. House of Representatives the same as it is now. To build the digital twin personas, we use the same pipeline and the same gender, education and ideology covariates as we used on our nuclear power simulation; we remove the level of environmental movement support covariate; and add a race covariate, a need for cognition score, and a vertical collectivism score which denotes a focus on group loyalty combined with acceptance of hierarchical structures.[24] As background information, we also include in the prompt "To help you choose an option, please know that expanding the size of the House will reduce the number of constituents in each district, enabling representatives to better connect with their district residents. However, adding new seats to the House would increase the cost of government and potentially make Congress more unwieldy."

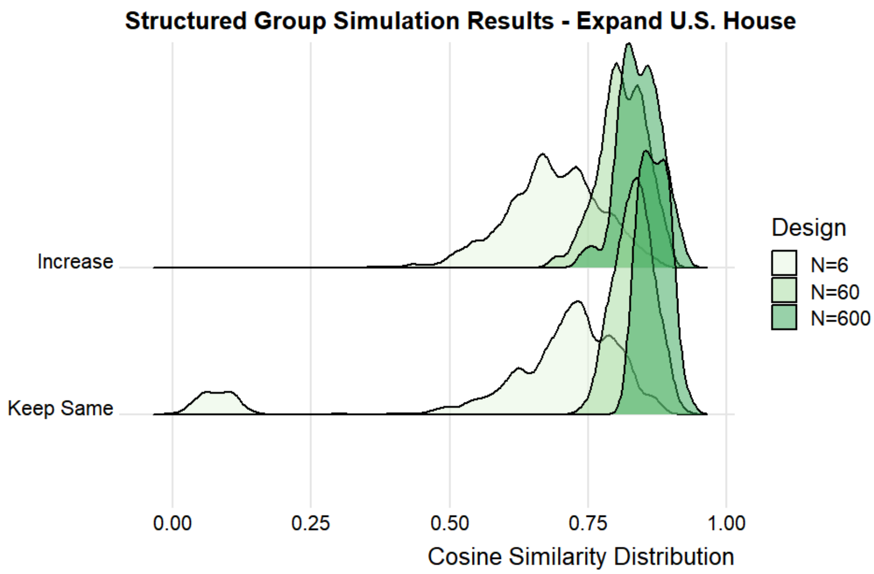


Figure A1: Distribution of pairwise cosine similarity scores, comparing between the small-$N$ focus group design and the large-$N$ design – two response options. Simulated focus group results on the topic of expanding the size of the U.S. House of Representatives. Response 1: In favor of expansion. Response 2: Opposed to expansion.

The figure A1 shows the distributions of cosine similarity scores across trials for each response. Notice that even with only two responses, there are many sessions where the small-$N$ six person focus group never discusses the option to keep the House the same size. These are groups where there is no disagreement. And as before, when a topic is discussed, the discussion shows considerable variability in the cosine similarity scores, indicating that each event produces relatively idiosyncratic discussion and data.

Figure A2 shows the bias audit results for the simulated data. Notice the strong simi-

[24]We add the need for cognition and vertical collectivism scores to the personas in order to ensure diverse responses to the prompt. In the absence of these covariates, liberal LLMs almost always chose to expand and conservative LLMs almost always chose to keep the House the same.

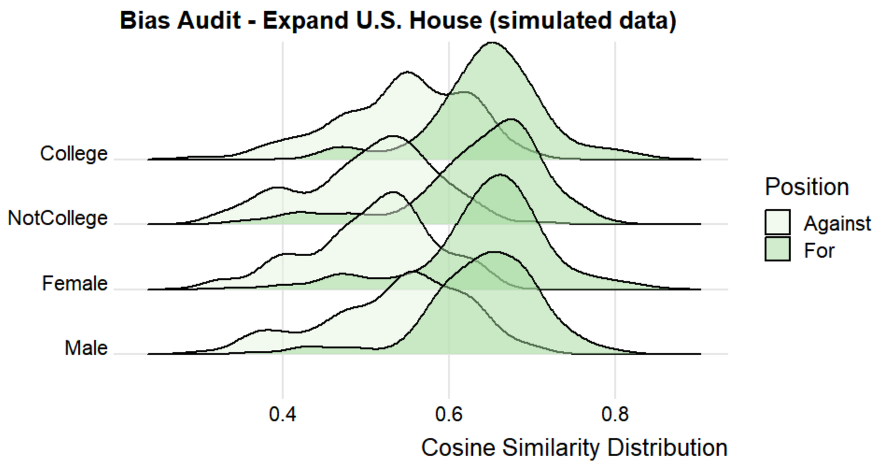


Figure A2: Distribution of simulated individual cosine similarity scores, conditional on demographics and position. $N$=600 design. Topic: Expansion of the U.S. House of Representatives

larities to the results of the bias audit between the data generated by humans as reported in figure 4 and in particular little evidence of bias in the summaries since the conditional distributions strongly overlap. The exception is that, as in the human data, college educated personas tended to have their responses more closely reflected in the summaries. Interestingly, even in the simulated data, supporters tend to have their arguments more closely reported in the summaries than opponents, which again suggests that there are more ways to oppose the policy than to support it. That the simulated data reflects these patterns helps to reinforce that the digital twin simulated arguments have good correspondence with human authored arguments.

**Bias Audit of the Summaries of Simulated Nuclear Power Responses.** Here for completeness we show the bias audit for our main simulation on the topic of nuclear power. To create the simulated audit data, we calculate the cosine similarity of each individual LLM-simulated persona's reasoning with the summary of the standpoint option they chose. Recall that we ask the summary tool to include five different perspectives within each summary. We calculate the cosine similarity between the simulated persona's response and each of the five summarized perspectives, and then retain the maximum value. That is, if simulated individual $i$ chose response option $k$, we calculated the cosine similarity between $i$'s response and the five response $k$ summarized perspectives, and then saved the maximum of these five values. Here we audit the summaries based on the stage 1 discussion.

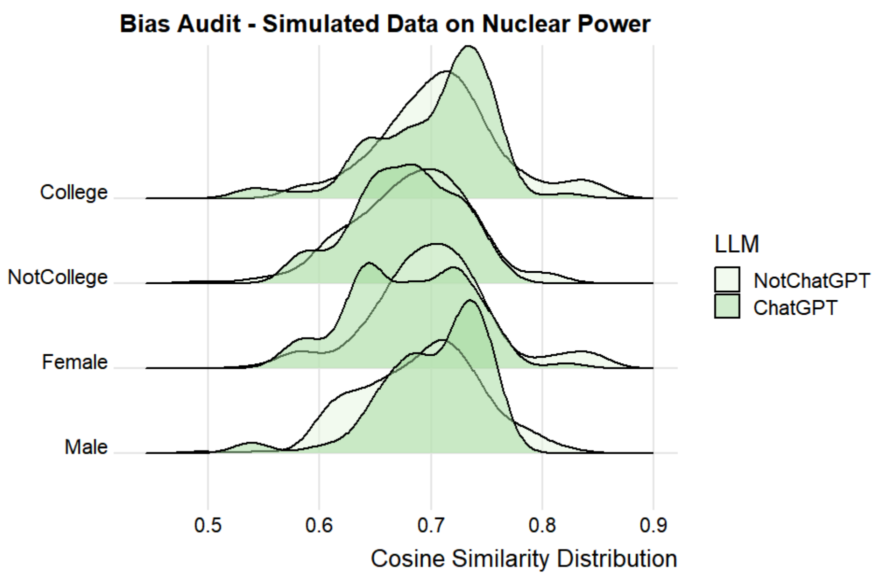


Figure A3: Distribution of simulated individual cosine similarity scores, conditional on demographics.

Figure A3 shows the conditional distributions of the cosine similarity scores for the simulated individuals, conditional on two covariates (education and gender) and the LLM (OSS-GPT versus the others). The conditional distributions indicate no systemic disparities by gender, or perhaps an under-representation of the female non-OSS-GPT perspectives. However, there is a slight over-representation of college-educated simulated individuals. As we show above, these results replicate in the human data. Finally, there is a noticeable over-representation of OpenAI's OSS-GPT, which is likely due to the fact that GPT provides more generic answers to the prompts.

Overall, in both the simulated and in the human-generated data, the conditional cosine similarity distributions in the bias tests are largely the same across all covariates. These results suggests that all diverse perspectives are reasonably well-captured and included in the summarized results.

# C Simulation Replicability and Reproducibility

Readers can run the simulations and validation scripts described in this paper by navigating to the project GitHub repository (URL omitted for review). Using the terminology from Brodeur et al. (2024), running the simulator with the script's preset parameter values will *replicate*, but not *reproduce*, the results we report in the paper. As Spirling (2023) notes, it is not possible to reproduce the simulation output since there is no way to set the seed for the LLM models we use. Further, Google and Groq deprecate or change the LLMs from time-to-time and we have no control over that. For these reasons, users cannot replicate the exact simulation results we report in the paper. However, since we focus on the distribution patterns of cosine similarity scores rather than point estimates for both the main results and the bias evaluation, the replicated results should be similar to the results we report in the paper. Indeed, Google discontinued the Gemini 2.0-flash model that we used in our initial experiments, but in our final experiments the results using Gemini 3.0-flash-preview were essentially identical in distribution. That we use multiple foundation models, instead of relying on only one, also mitigates these concerns.

We provide the data and code in our project repository (URL omitted for review) so that readers can reproduce the paper's figures using our simulated results. Our reproduction repository shows we can reproduce the results reported in the paper from the raw data that resulted from our simulation runs. Reproduction in this context is analogous to analyzing survey data provided by a vendor. It would be strange to require a researcher to re-administer a survey and expect the survey results to be identical, even with the same respondents. Likewise, our reproduction repository shows that we can reproduce the results in the paper given the data provided by our simulation runs.

# D AI Disclosure

Here we disclose the use of AI for this project.

**What AI was used for.** We used AI in three ways for this project. First, as we describe explicitly in the main text, we used LLMs in the simulation, analysis and validation pipelines. The simulator uses covariate data to construct LLM prompts, which in turn create an ensemble of LLM "personas" that are tasked with deliberating an issue. The simulator also uses an LLM prompt to summarize the content that the ensemble of personas generate. For the main analysis as well as our bias audit, we convert the LLM summarized text to embeddings in order to implement cosine similarity metrics. For validation, we make use of an LLM to classify persona-generated content by topic. Each aspect of the simulation, analysis and validation pipelines, and how we use LLMs for each, is explained in detail and explicitly in the manuscript.

Second, we made use of an LLM to streamline the Python script for the core structured group design simulator. To do this streamlining, we asked the LLM to read the human-written version of the simulator, and we asked it to create a new version of the code that works (exactly) identically but is more efficiently coded. All of the underlying logic of the simulator remains human-written, but the code in the updated version is a bit more modular (and the LLM had a preference for using data classes for storage instead of dictionaries that we had used). We disclose this use of the LLM in the acknowledgments section of the paper and in the code repository.

Third, as we describe explicitly in section 11 of the paper, we made use of an LLM to create a new Python script for the unstructured group design. As we note in the paper, using the LLM to create this new script was an experiment to demonstrate that the code repository is easily extensible to other synchronous designs. We only report that we conducted this experiment in code generation but we do not report results from this LLM-generated simulator.

**Which tools were used.** To generate simulated data among our LLM personas, we make use of four different LLMs: Gemini 3.0 flash preview, Qwen3 32B, Llama 3.3 70B versatile, and OpenAI GPT OSS 120B, with the latter three being hosted on Groq and are open weight LLMs. To generate the summaries we use Gemini 3.0 flash preview. We state these models in footnote 12. To create embeddings for the analysis, we use the Hugging Face sentence transformer all-MiniLM-L6-v2. We state this model in footnote 11. To do the coding streamline project and the new script generation, we use Claude Sonnet 4.6. We also state this model in the acknowledgment section.

**How human oversight was maintained.** A core aspect of our paper is the development and implementation of validation strategies for LLM-based discourse simulation. We describe our validation methods in section 4 of the manuscript, including our proposals for backward continuity, frontward continuity and coverage, each of which required a research assistant to review the LLM output for accuracy, and then the full lab met to review the research assistant's work. We report the results of the validation tests in section 8 of the manuscript. We also implement a bias audit of the LLM summaries that relies on human-generated data for the assessment, and we report the results of this audit in section 9 of the manuscript. Finally, for the code streamlining task and the new script task, the PI read the code line by line to ensure that code was exact and correct.

Table A1: Summaries for Nuclear Power Focus Group

| | |
|---|---|
| | Discussion Options:<br>1 = Implement rapid expansion and investment of nuclear power<br>2 = Maintain the current nuclear energy operations without change<br>3 = Phase out existing nuclear plants and halt new construction |
| **Stage 1: Initial Arguments** | |
| Option 1 | Nuclear power is viewed as a reliable, low-carbon energy source necessary for meeting climate goals and reducing reliance on fossil fuels. Energy independence and security are seen as benefits of nuclear power, reducing dependence on foreign sources and unreliable renewables. Technological advancements in safety and waste management are believed to mitigate concerns associated with nuclear power. Nuclear power is considered a viable solution for meeting increasing energy demands while providing economic benefits such as job creation. Nuclear power is seen as a practical solution for transitioning away from fossil fuels, especially when considering the limitations of intermittent renewable energy sources. |
| Option 2 | A balanced approach is favored, as it allows for the continued use of a significant energy source without the perceived risks of rapid expansion or the negative impacts of premature phase-out. The status quo is seen as a pragmatic solution, given existing political, social, and environmental complexities, as well as the desire to avoid further polarization while utilizing current infrastructure. Concerns regarding the risks associated with nuclear waste and potential accidents are considered significant enough to warrant caution, emphasizing the need for further advancements in safety and waste management before any expansion is considered. Maintaining current nuclear operations allows for a reliable, low-carbon energy source to be preserved, while further research into safer and more sustainable energy solutions is conducted. The current approach is viewed as a responsible middle ground, balancing energy needs with environmental and safety concerns, and allowing for continued research and public discourse before committing to significant changes. |
| Option 3 | The dangers of long-term nuclear waste storage and the potential for catastrophic accidents are considered unacceptable risks. Investment in renewable energy sources is believed to be a more environmentally responsible and sustainable alternative. The risks associated with nuclear energy are viewed as outweighing any low-carbon benefits. Concerns about the safety and potential environmental impact of nuclear energy are prioritized over its advantages. |

Table A1: Summaries for Nuclear Power Focus Group (cont.)

| | |
|---|---|
| | Discussion Options:<br>1 = Implement rapid expansion and investment of nuclear power<br>2 = Maintain the current nuclear energy operations without change<br>3 = Phase out existing nuclear plants and halt new construction |
| **Stage 2: Counterarguments** | |
| Option 1 | Long-term safety and waste disposal concerns are seen as unresolved, with the potential for accidents and mismanagement outweighing the benefits, regardless of technological advancements. The high costs, long construction times, and complex supply chains associated with nuclear power are viewed as potential impediments to a rapid energy transition, diverting resources from more readily available and resilient renewable solutions. The necessity of overhauling the existing energy infrastructure to prioritize nuclear power is questioned, particularly when current needs are being met and other renewable options are available. Prioritizing nuclear power for energy independence is believed to create new vulnerabilities and distract from investments in domestic renewable energy sources. The potential long-term risks associated with rapid nuclear expansion, including waste disposal and safety concerns, are considered to outweigh the economic benefits of job creation and meeting energy demands. |
| Option 2 | A reliance on fossil fuels is seen to be prolonged by not expanding nuclear power, potentially missing critical climate targets due to the limitations of renewable energy sources alone. The risks associated with nuclear power, such as waste disposal and potential accidents, are considered to outweigh the short-term benefits, with a preference for prioritizing a rapid transition to renewable energy sources. A delay in scaling up nuclear power is viewed as a risk to energy security and climate action, as nuclear is currently seen as a readily available solution, and scaling up is preferred while pursuing other innovations. Maintaining the current status of nuclear operations is considered insufficient for achieving necessary emission reductions, and a more proactive approach involving new construction is seen as necessary to displace fossil fuels and meet energy demands. Continuing current nuclear operations is seen as delaying the transition to truly sustainable and secure energy solutions, with concerns that it perpetuates reliance on outdated systems and diverts resources from renewable alternatives. |
| Option 3 | The reliable baseload power provided by nuclear energy is considered essential for grid stability and meeting energy demands, especially when renewables are intermittent or not yet scalable. Modern advancements in nuclear technology and stringent safety protocols are believed to have significantly mitigated risks associated with nuclear energy, making it |

Table A1: Summaries for Nuclear Power Focus Group (cont.)

| | Discussion Options:<br>1 = Implement rapid expansion and investment of nuclear power<br>2 = Maintain the current nuclear energy operations without change<br>3 = Phase out existing nuclear plants and halt new construction |
|---|---|
| (cont.) | a safe and manageable energy source. The risks associated with nuclear energy, including waste management and potential accidents, are viewed as manageable with proper technology and regulation, and are outweighed by the existential threat of climate change and the harms of fossil fuels. Nuclear energy is seen as a critical component in transitioning to a low-carbon economy and achieving energy independence, offering a proven, scalable, and low-carbon alternative to fossil fuels without the ecological trade-offs of renewables. Prematurely phasing out existing nuclear plants is viewed as counterproductive, potentially leading to increased reliance on fossil fuels and hindering progress towards climate goals, as well as wasting existing infrastructure. |
| **Stage 3: Responses to Counterarguments** | |
| Option 1 | Nuclear power is believed to be a necessary component of a diversified energy portfolio, offering a reliable baseload power source that complements intermittent renewable energy sources and reduces reliance on fossil fuels. Concerns about the safety and waste disposal associated with nuclear power are acknowledged, but it is argued that advancements in reactor designs, waste management technologies, and regulatory oversight have significantly mitigated these risks, making them manageable. The expansion of nuclear power is considered an investment in long-term energy independence, providing a stable and secure energy supply that reduces vulnerability to volatile global markets and geopolitical instability. The urgency of addressing climate change is emphasized, with nuclear power viewed as a crucial tool for decarbonizing the energy sector quickly and effectively, especially given the limitations of renewables in meeting baseload power demands. Modernizing infrastructure and investing in nuclear power is seen as a means of future-proofing the energy grid, preparing for increased energy demands and ensuring a sustainable energy supply for future generations. |

Table A1: Summaries for Nuclear Power Focus Group (cont.)

| | |
|---|---|
| | Discussion Options:<br>1 = Implement rapid expansion and investment of nuclear power<br>2 = Maintain the current nuclear energy operations without change<br>3 = Phase out existing nuclear plants and halt new construction |
| **Stage 3: Responses to Counterarguments (cont.)** | |
| Option 2 | A rapid expansion of nuclear power is viewed as unnecessary, as renewable energy sources are believed to have the potential for further development and could meet energy needs. Maintaining current nuclear operations is seen as a pragmatic approach to ensure a stable energy supply while improvements to waste management and safety technologies are made and renewables are scaled up. Decommissioning existing nuclear plants before renewables are fully capable of meeting demand is viewed as risky, potentially creating energy instability and increasing reliance on fossil fuels. Building new nuclear plants is seen as a potentially risky and costly endeavor, with concerns raised about waste disposal, long-term risks, and the prioritization of renewable energy sources. A cautious approach to nuclear energy is favored, focusing on optimizing existing plants, improving safety, and exploring the potential of renewable energy sources before committing to large-scale new construction. |
| Option 3 | The long-term risks associated with nuclear power, such as waste disposal and potential accidents, are believed to outweigh the benefits, especially when renewable energy sources are available. Reliance on nuclear power is viewed as unnecessary due to the availability of cleaner, safer, and more sustainable alternatives like solar and wind power. Concerns are raised that the term manageable does not equate to safe, particularly in the context of nuclear power's potential for catastrophic accidents and long-term environmental impact. The possibility of a short-term energy gap resulting from phasing out nuclear plants is acknowledged, but it is suggested that this gap can be bridged with investments in renewable energy and storage solutions. Continued investment in nuclear power is seen as diverting resources from the development and deployment of renewable energy technologies. |